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Muscular system

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Introduction

1. Definition of Muscle

The muscle, the driving force of the locomotor system, is an organ that contracts (shortens) under the influence of a stimulus (nerve, electrical, thermal, painful excitation, etc.). The study of muscles is Myology.

2. Roles of the muscular system

- Production of the movement
- Heat production (thermogenesis)
- Maintaining posture
- Joint stabilization
- Protection of the viscera
- Visceral functions : Digestion, Defecation, Urination, Blood propulsion, Vasoconstriction, Vasodilation, Childbirth, etc.

3. Properties of muscles

- Excitability and Contractility: it is the ability to contract following an excitation;
- Elasticity: it is the ability to stretch (without being torn);
- Tone: This is the ability to maintain a slight, permanent muscle contraction, despite its relaxation. It allows us to maintain balance at rest (posture), and disappears upon lying down, in a coma, and in death (atonic subject).

4. Muscle classification

There are three types of muscles…

	Skeletal Muscle	Cardiac Muscle	Visceral muscle
Other names	Striped, Red	Mixed	Smooth
Cytology	Striated cells	Striated cells	Non-striated cells
Location	Attached to the skeleton	Heart walls	hollow viscera Ship walls Skin
Muscle contraction	Voluntary Fast	Involuntary Automatic Rhythmic Fast	Involuntary Slow Continue
Fatigability	Fast	Never	Never
Nervous control by :	Central nervous system (CNS)	Autonomic nervous system (ANS)	Autonomic nervous system (ANS)

Skeletal muscle

1. Number

There are : 600 constant skeletal muscles.

This number varies due to the existence of inconstant muscles, which are accessory muscles (Ex : plantaris muscle).

2. Situation

Two muscle groups :

- Deep muscles (sub-aponeurotic). Ex: muscles attached to the digestive and respiratory systems, skeletal muscles, muscles attached to the sensory organs;
- Superficial muscles (supra-aponeurotic, subcutaneous), Ex: facial muscles.

3. Division

A simple muscle is a monogastric muscle, composed of three parts:

- A middle section : called the muscular belly (or body), it is large, red, fleshy and contractile;
- Two ends : called muscle tendons, they are narrow, whitish, elastic and resistant.

Besides this division, muscle can present four varieties:

- Digastric muscle: it is a muscle which has two bellies and an intermediate tendon. Ex : digastric muscle;
- Multigastric muscle: it is a muscle that has several bellies and several intermediate tendons. Ex: rectus abdominis muscle;
- Bifid muscle: it is a muscle that has two bellies that are arranged in parallel. Ex : biceps brachii muscle;
- Multifidus muscle: It is a muscle that has several bellies arranged in parallel. Ex : quadriceps femoris muscle.

Noticed: In the case of a bifid or multifid muscle, the parallel tendons are called heads.

4. Insertion

The muscle attaches via a double muscular insertion:

- Original Insertion, called proximal or fixed;
- Inserting the termination, called distal or mobile.

Noticed: Sometimes, the original insertion can become mobile.

Muscle insertion can occur at four sites:

- Bone insertion. Ex : iliopsoas muscle ;
- Cartilage insertion. Ex : cricothyroid muscle ;
- Aponeurotic insertion. Ex: abductor hallucis muscle;
- Skin insertion. Ex : palmaris brevis muscle.

5. Shape

- Long muscle: its length predominates over its width and thickness. Ex: sartorius muscle;
- Flat muscle: its length and width predominate over its thickness. Ex : latissimus dorsi muscle ;
- Short muscle: the three dimensions are reduced. Ex: muscles of the hypothenar compartment;
- Annular muscle: is located around the openings. Ex: orbicularis oculi muscle of the eyelids, orbicularis oris muscle of the lips;
- Geometric muscle : Ex : trapezius muscle, pronator quadratus muscle, depressor anguli oris muscle.

6. Summary : Nomenclature

Muscle nomenclature is based on morphological and functional criteria.

- Size criterion. Ex : gluteus maximus muscle, pectoralis minor muscle, fibularis longus muscle ;
- Action criterion. Ex: adductor magnus muscle, supinator muscle, superficial flexor muscle of the fingers;
- Situational criterion. Ex : dorsal interosseous muscles, gluteal muscles, brachialis muscle;
- Form criterion. Ex : piriformis muscle, quadratus plantaris muscle ;
- Muscle fiber direction criterion. Ex : internal oblique muscle, rectus abdominis muscle ;
- Insertion criteria. Ex : brachioradialis muscle, coracobrachialis muscle ;
- Number of heads criterion. Ex: biceps femoris muscle, triceps brachii muscle.

7. Appendices

These are anatomical structures that are useful to the muscle, facilitating its action. There are four:

- Serous (synovial) bursae: these are membranous sacs filled with synovial fluid, located near the joints and facilitate the gliding of muscles;
- Tendinous serous sheaths: these are serous bursae surrounding the tendons;
- Fibrous sheaths: these are fibrous grooves, which delimit with the bone an osteofibrous canal, where the tendons slide, enveloped by serous sheaths;
- Muscular fascia (covering aponeuroses): These are fibrous membranes, that encase and separate the muscles.